Improving Diet and Fitness Coaching with an Intelligent Mobile Application

Mehmet Can Irmak – 210444076
Can Acar – 200444037

1 - Introduction

1.1. Purpose of the System

This system's main goal is to revolutionize the interaction and monitoring processes between experts (dietitians and personal trainers) and their clients. Manual reporting is frequently used in traditional diet and activity tracking systems, which can be time-consuming, inaccurate, and ineffective for both parties. By utilizing mobile technology and artificial intelligence, this application seeks to overcome these constraints and offer a more efficient, data-driven, and customized experience.

1.2. Scope of the System

The initial scope of the system will focus on the following key features:

- **AI-Powered Food Image Analysis:** The application will use a machine learning model (e.g., YOLO) to analyze photos of food taken by users. It will identify food items and estimate their calorie content, providing data to dietitians.
- **Temporary Exercise Video Sharing:** Clients can record and share short videos of their workouts with their personal trainers. These videos will be viewable for a limited time to prevent data overload on the trainer's device.
- **Expert-Client Communication:** The application will provide a communication channel for experts and clients to exchange messages and feedback.
- **Data Visualization for Experts:** The application will present analyzed dietary data in a user-friendly format for experts to review and use for making informed recommendations.

1.3. Objectives and Success Criteria of the Project

The primary objectives of this project are:

- **Develop a Functional Mobile Application:** Create a fully functional mobile application for both iOS and Android platforms, containing the basic features described.
- **Achieve Accurate Food Image Analysis:** Train and implement the AI model to achieve a high level of accuracy in identifying food items and estimating calorie content. A target accuracy of 85% will be considered successful.
- **Ensure Seamless Video Sharing:** Implement a stable and efficient video sharing mechanism that allows for temporary viewing and prevents unnecessary files.
- **Provide a User-Friendly Interface:** Design an user-friendly interface that is easy to navigate and use for both experts and clients.

1.4. Definitions, Acronyms, and Abbreviations

Definitions:

- **AI:** Artificial Intelligence
- **YOLO:** You Only Look Once (object detection model)
- **CNN:** Convolutional Neural Network - A type of neural network commonly used for image processing and analysis.
- **HCI:** Human-Computer Interaction - A field of study focused on the design and use of computer technology, focusing on the interfaces between people (users) and computers.
- **UI:** User Interface - The means by which a user interacts with a computer or software application.

Acronyms:

- **BMI:** Body Mass Index
- **BFP:** Body Fat Percentage
- **CHO:** Carbohydrates
- **P:** Protein
- **F:** Fat

1.5. References

- **Original YOLO Paper:** Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. *Proceedings of the IEEE conference on computer vision and pattern recognition*, 779-788.
- **Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT press.** (A comprehensive textbook on deep learning)
- **Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25.** (A foundational paper on CNNs)
- **Apple Developer Documentation:** https://developer.apple.com/ (Official documentation for iOS development)
- Ultralytics. (n.d.). *YOLOv8 Documentation*. Retrieved from https://docs.ultralytics.com/

2 - Current System

There are a lot of health and fitness apps on the market and they generally focus on nutrition tracking, exercise planning or weight tracking as isolated functions. Unlike these common apps, we will create a communication and analysis app that emphasizes the relationship between experts and their clients.

2.1 Limitations of Existing Systems

Manual Data Entry: Many applications require users to manually input the food they consume, including portion sizes and calorie estimates. This process is not only time-consuming but also prone to user error, leading to inaccurate tracking.

Lack of Real-Time Food Recognition: Most current systems lack advanced tools like AI-powered visual recognition to automatically identify food items and calculate their nutritional values.

Fragmented Features: While some applications focus only on features such as fitness tracking diet planning, we blend these features and offer expert client communication and analysis together.

Absence of Smart Recommendations: Applications where real-time data provided by the user can be analyzed in detail and presented to experts for smart recommendations and planning are not common.

Minimal Integration with Smart Devices: Although some apps connect to fitness trackers, they often fail to utilize real-time data from these devices to offer actionable insights or feedback

2.2 User Problems in Existing Systems

• **Difficulty in Meal Logging:** Users find it tedious to manually log meals and input food details every time.

• **Lack of Engagement:** Without real-time tracking and interactive features, users often lose motivation and stop using the applications over time.

• **Inaccurate Progress Tracking:** Applications relying on user-entered data often result in incomplete or inaccurate health tracking, failing to provide realistic insights.

•**Time-Consuming Processes:** The manual processes in current systems, such as searching food items, inputting calories, and setting up plans, are inefficient for users seeking quick and automated solutions.

Proposed System

3.1 Overview

The Health & Fitness Expert Communication Platform is a unique mobile application that upgrades the communication and follow-up process between health experts, including dietitians and personal trainers, and their clients. This system will try to simplify the consultation process and enhance the efficacy of health and fitness programs by using artificial intelligence and modern communication technologies.

The platform solves some of the main problems of the current health and fitness consultation industry: ineffective means of communication, a burden of manual data entry, and disorganized tracking processes. The application was designed to help experts and clients in a much more efficient and user-friendly way, introducing AI-powered features and temporary content-sharing mechanisms.

3.2 Functional Requirements:

Functional requirements describe *what* the system should do. They define specific actions or functions that the system must perform.

- **Food Image Analysis:**
    - The system shall allow users (clients) to capture images of their meals using their mobile device's camera.

    - The system shall analyze the captured food images using an AI model (e.g., YOLO) to identify the food items present.
    - The system shall estimate the approximate calorie content of the identified food items.
    - The system shall provide the analysis results (food items and estimated calories) to the dietitian.
- **Exercise Video Sharing:**
    - The system shall allow users (clients) to record short videos of their exercises.
    - The system shall allow users (clients) to send these videos to their personal trainer in a temporary/ephemeral format (similar to "snaps").
    - The system shall ensure that these videos are automatically deleted from the trainer's device after a defined period (e.g., 24 hours).
- **Communication & Notification:**
    - The system shall notify dietitians of newly analyzed food images.
    - The system shall notify personal trainers of received exercise videos.
    - The system shall facilitate direct communication (e.g., messaging) between clients and their respective experts (dietitians/trainers).
- **User Authentication & Authorization:**
    - The system shall allow users (both clients and experts) to create accounts and log in securely.
    - The system shall enforce role-based access control, ensuring that clients can only access their own data and experts can only access data of their assigned clients.
- **Data Storage and Management:**
    - The system shall securely store user data, including personal information, food logs, exercise videos (temporarily), and communication history.
    - The system should allow experts to view historical data of their clients.

3.3 Non-Functional Requirements:

Non-functional requirements describe *how* the system should perform. They define the quality attributes, constraints, and operational characteristics of the system.

- **Performance:**
    - **Image Analysis Speed:** The system shall analyze food images and provide results within a reasonable timeframe (e.g., under 5 seconds).
    - **Video Upload/Download Speed:** The system shall allow for efficient uploading and downloading of exercise videos, considering varying network conditions.
    - **System Responsiveness:** The application should be responsive to user interactions, with minimal delays.
- **Security:**
    - **Data Confidentiality:** The system shall protect user data from unauthorized access, disclosure, alteration, and destruction. This includes encryption of sensitive data both in transit and at rest.
    - **Authentication Security:** The system shall use secure authentication mechanisms (e.g., strong passwords, multi-factor authentication) to prevent unauthorized logins.
    - Video and image security: The system must ensure that shared media is only accessible to the intended recipient and for the intended duration.
- **Usability:**
    - **Intuitive Interface:** The application shall have a user-friendly and intuitive interface that is easy to navigate and understand for both clients and experts.
    - **Accessibility:** The application should adhere to accessibility guidelines to accommodate users with disabilities.
    - **Clear Instructions:** Provide clear instructions and help documentation within the application.
- **Scalability:**

    - The system shall be able to handle a growing number of users and data without significant performance degradation. The system should be designed to accommodate future expansion.

- **Reliability:**
    - The system shall be available and operational for a high percentage of time (e.g., 99.9% uptime).
    - The system shall be robust and resilient to errors and failures.
- **Maintainability:**
    - The system's code should be well-structured, documented, and easy to understand, making it easy to maintain and update.
- **Portability:**
    - The application should be compatible with different mobile operating systems (e.g., iOS and Android).
- **Constraints:**
    - **Development Time:** The project must be completed within the allocated timeframe for the graduation project.
    - **Budget:** The project must be developed within a reasonable budget (if applicable).
    - **Technology:** The project must use specified or available technologies and libraries.

3.4 System Models

3.4.1 Scenarios

These scenarios demonstrate the system's capability to handle various situations while maintaining data integrity, user privacy, and service quality. They also highlight the platform's flexibility in adapting to different user needs and unexpected circumstances.

Scenario 1: Daily Meal Logging and Analysis

User: Sarah (Client)

Expert: Dr. Ayşe (Dietitian)

Flow:

1. Sarah has lunch at a restaurant and takes a photo of her meal through the app
2. The AI system analyzes the image and identifies:
   - Grilled chicken breast (150g)
   - Brown rice (100g)
   - Mixed vegetables (120g)
3. The system calculates approximate calories (520 kcal) and macronutrients
4. Dr. Ayşe receives a notification and reviews the meal
5. Dr. Ayşe notices the portion size is larger than Sarah's plan and sends feedback
6. Sarah receives the feedback immediately and adjusts her dinner portions accordingly

Scenario 2: Exercise Form Check

User: Mehmet (Client)

Expert: Ali (Personal Trainer)

Flow:

1. Mehmet records a 30-second video of his squat form during his workout
2. He uploads it through the temporary sharing feature
3. Ali receives a notification and reviews the video
4. Ali notices Mehmet's knees caving in and provides feedback with timestamp markers
5. Mehmet receives the feedback and corrects his form in real-time
6. The video automatically deletes after 24 hours

Complex Usage Scenarios

Scenario 3: Collaborative Care

User: Zeynep (Client)

Experts: Dr. Deniz (Dietitian), Selin (Personal Trainer)

Flow:

1. Zeynep has both a nutrition and exercise plan
2. Dr. Deniz updates Zeynep's meal plan due to recent blood test results
3. The system automatically notifies Selin about the dietary changes
4. Selin adjusts Zeynep's workout intensity accordingly
5. Both experts can view shared progress metrics
6. Zeynep receives coordinated updates from both experts

Scenario 4: Progress Tracking and Goal Adjustment

User: Can (Client)

Expert: Elif (Personal Trainer)

Flow:

1. Can logs his workouts and measurements for 4 weeks
2. The system generates a progress report showing:
   - Weight plateaued in week 3
   - Strength increases slowing down
   - Consistency in workout attendance
3. Elif reviews the data and identifies potential overtraining
4. She adjusts Can's program through the platform
5. The system automatically updates Can's workout schedule
6. Can receives notifications about the program changes

Edge Cases and Unexpected Situations

Scenario 5: Connection Issues

User: Aylin (Client)

Expert: Dr. Murat (Dietitian)

Flow:

1. Aylin tries to upload a meal photo in an area with poor connectivity
2. The app stores the photo locally and shows a pending status
3. The system automatically retries upload when connection improves
4. Dr. Murat receives the delayed upload with original timestamp
5. The system notifies Aylin of successful upload and expert review
6. All related notifications and data sync properly once connection restored

Scenario 6: Multiple Device Usage

User: Berk (Client)

Expert: Canan (Personal Trainer)

Flow:

1. Berk starts logging his workout on his phone
2. His phone battery dies mid-session
3. He switches to his tablet to continue the session
4. The system seamlessly syncs his partial workout data
5. Canan receives the complete workout log
6. All progress tracking remains accurate across devices

Scenario 7: Content Moderation

User: Emre (Client)

Expert: Leyla (Dietitian)

Flow:

1. Emre accidentally uploads a personal photo instead of his meal
2. The AI system flags the content as non-food
3. The upload is automatically rejected
4. Emre receives a notification explaining the issue
5. The system guides him to retry with appropriate content
6. The incorrect photo is immediately deleted from the system

System Recovery Scenarios

Scenario 8: Data Recovery

User: Merve (Client)

Expert: Burak (Personal Trainer)

Flow:

1. Merve's phone crashes while uploading a workout video
2. The app creates a local backup of the partial upload
3. Upon reopening, the app offers to resume the upload
4. The system verifies the video integrity
5. Burak receives the complete video with no quality loss
6. The system maintains accurate timestamps despite the delay

Scenario 9: Expert Unavailability

User: Deniz (Client)

Expert: Dr. Sevgi (Dietitian)

Flow:

1. Dr. Sevgi sets her status to "Away" for a week-long conference

2. The system automatically notifies all her clients

3. Urgent notifications are routed to her designated backup

4. The AI continues to analyze and log client meals

5. All data is queued for her review upon return

6. Clients receive automated responses with expected review timelines

Administrative Scenarios

Scenario 10: Expert Account Management

Administrator: System Admin

Expert: Yasemin (New Personal Trainer)

Flow:

1. Yasemin completes the expert registration process

2. Admin verifies her credentials and certifications

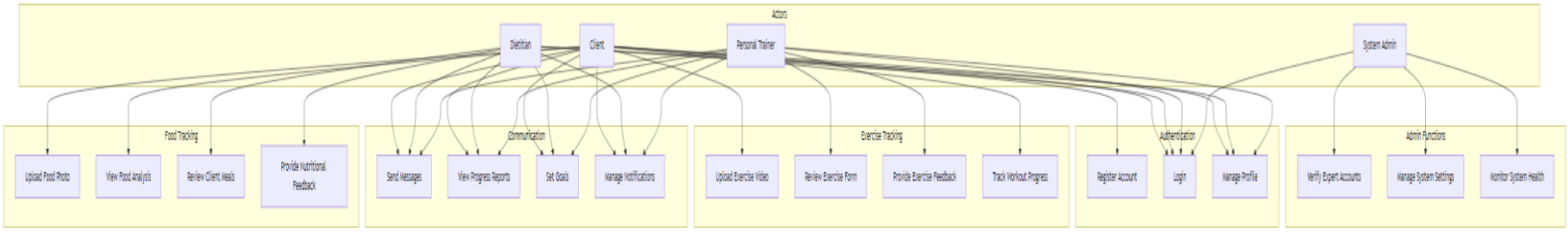
3. The system creates her professional profile

4. She receives access to expert-only features

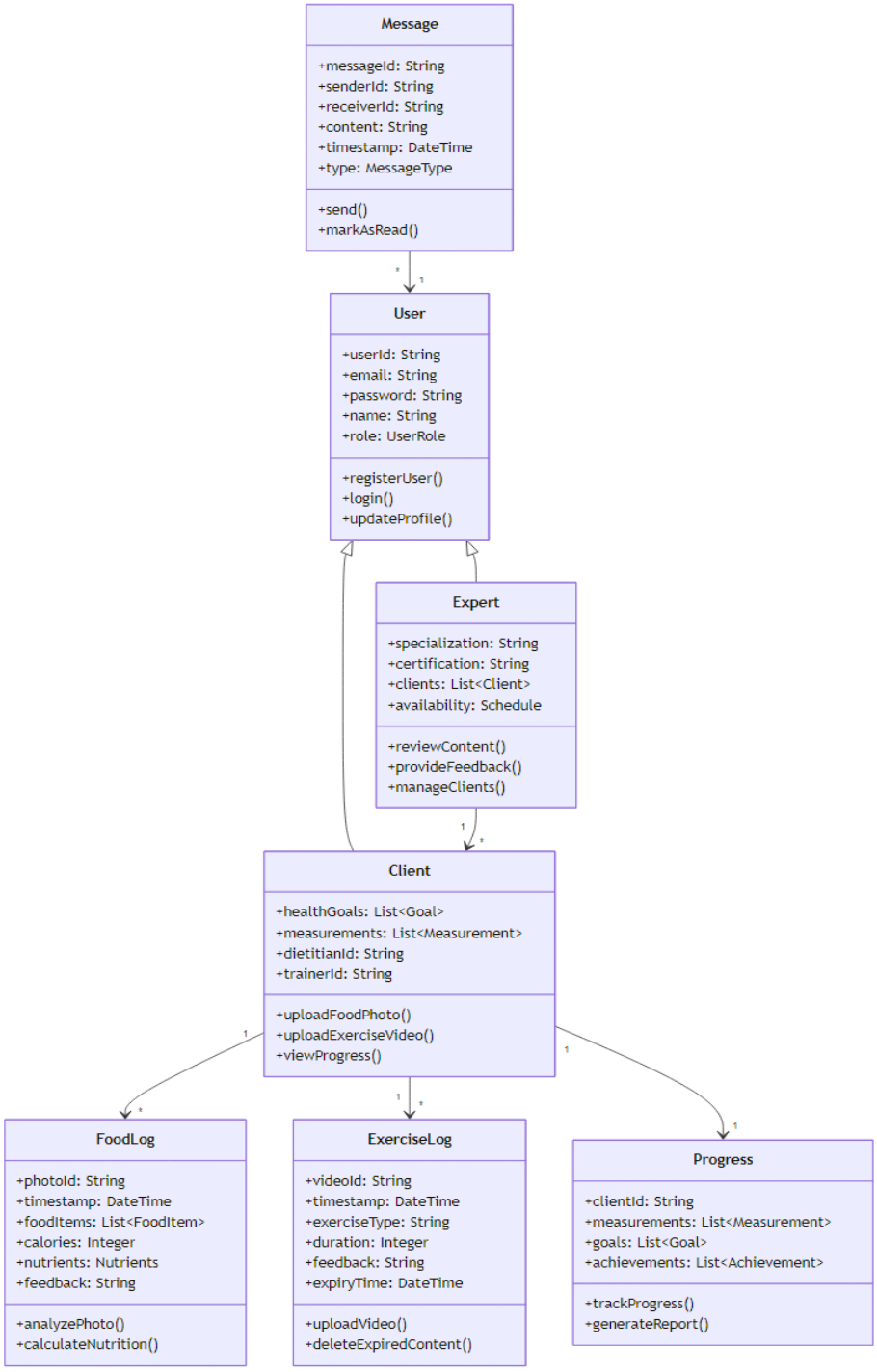
5. The system guides her through a mandatory training module

6. Her profile becomes visible to potential clients after completion

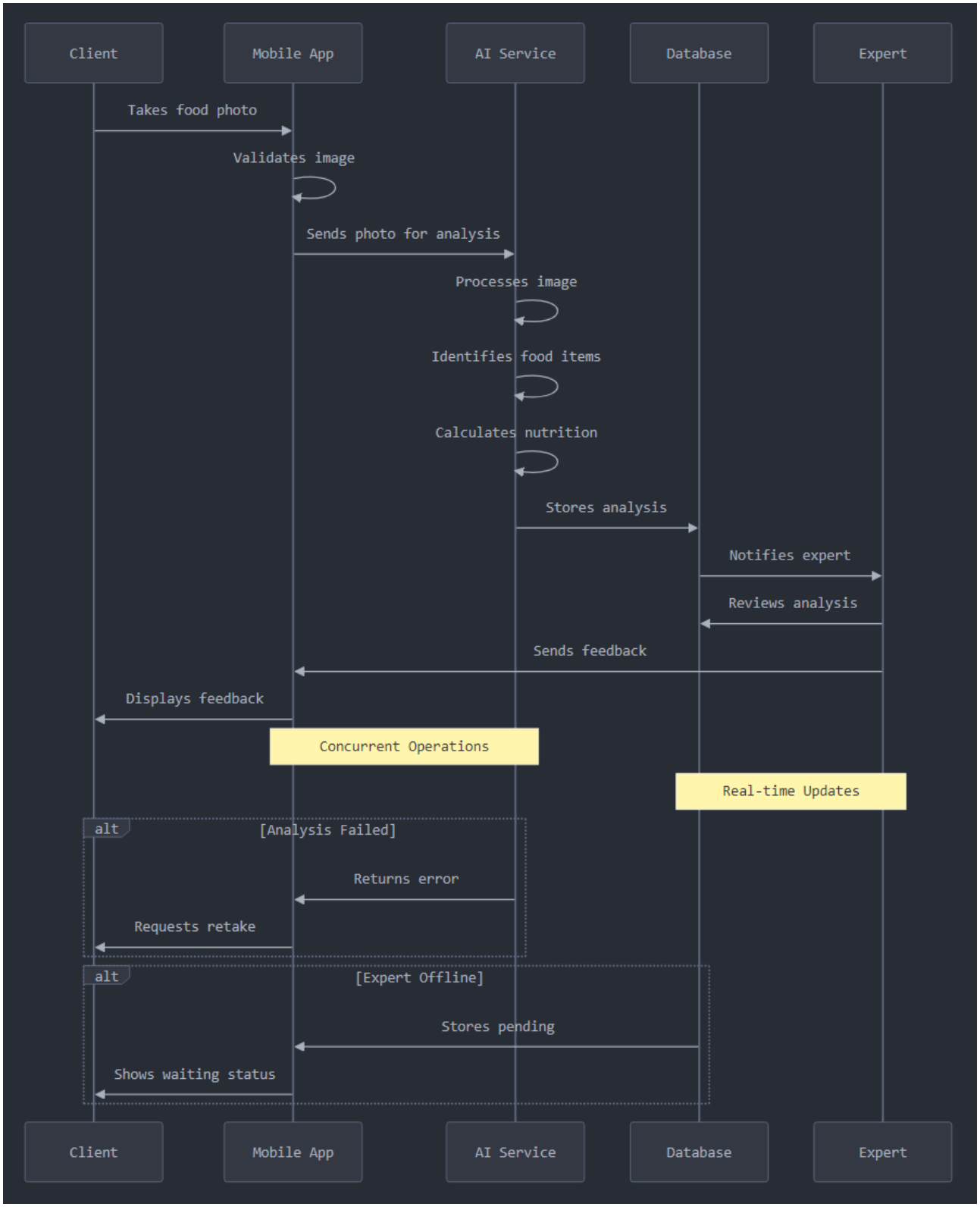
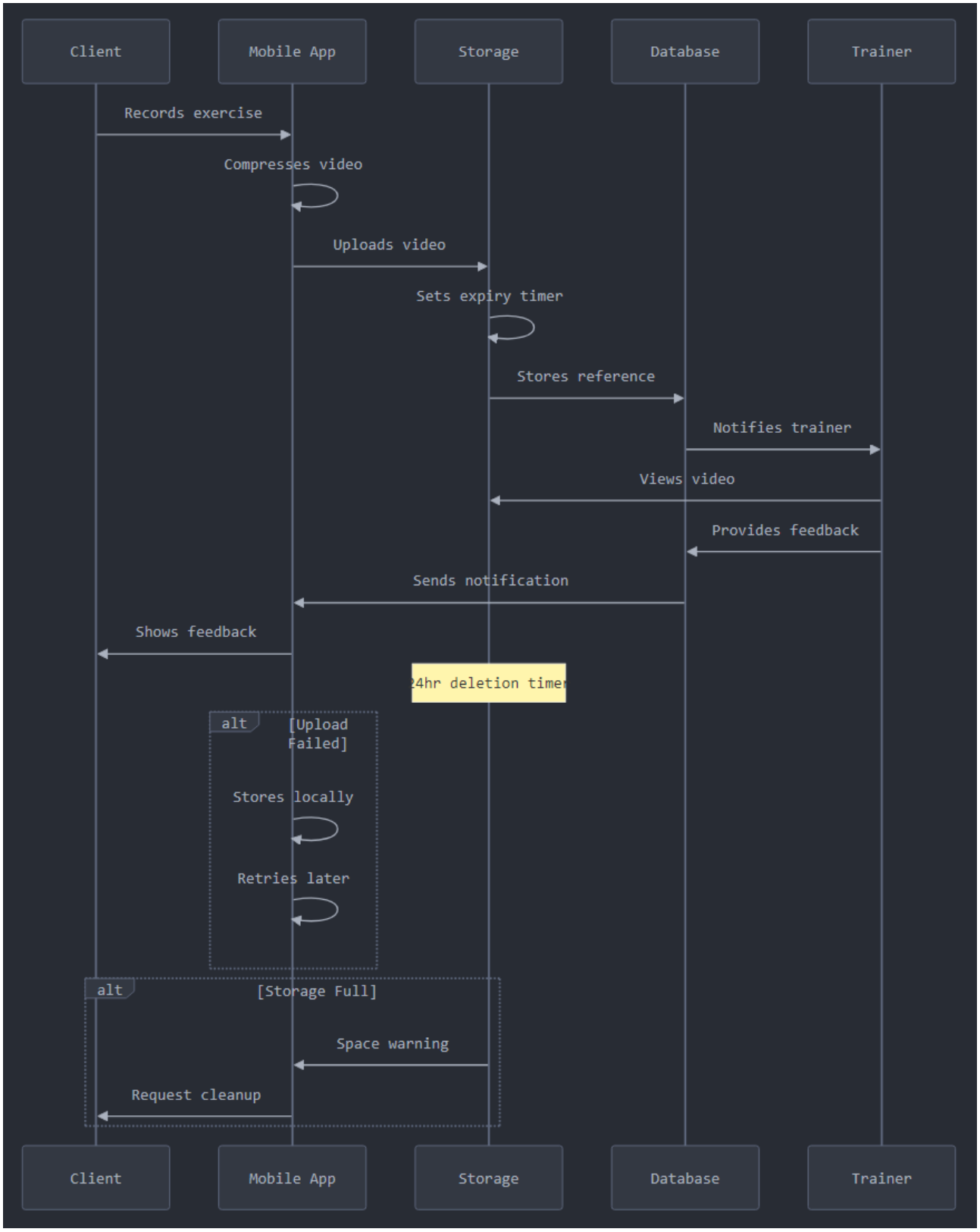
3.4.2 Use Case Model



3.4.3 Analysis Object Model



3.4.4.a Dynamic Model - Food Analysis Sequence



3.4.4.b Dynamic Model – Exercise Video Sequence